Introduction to Machine Learning

Dr Ahmed J. Aljaaf

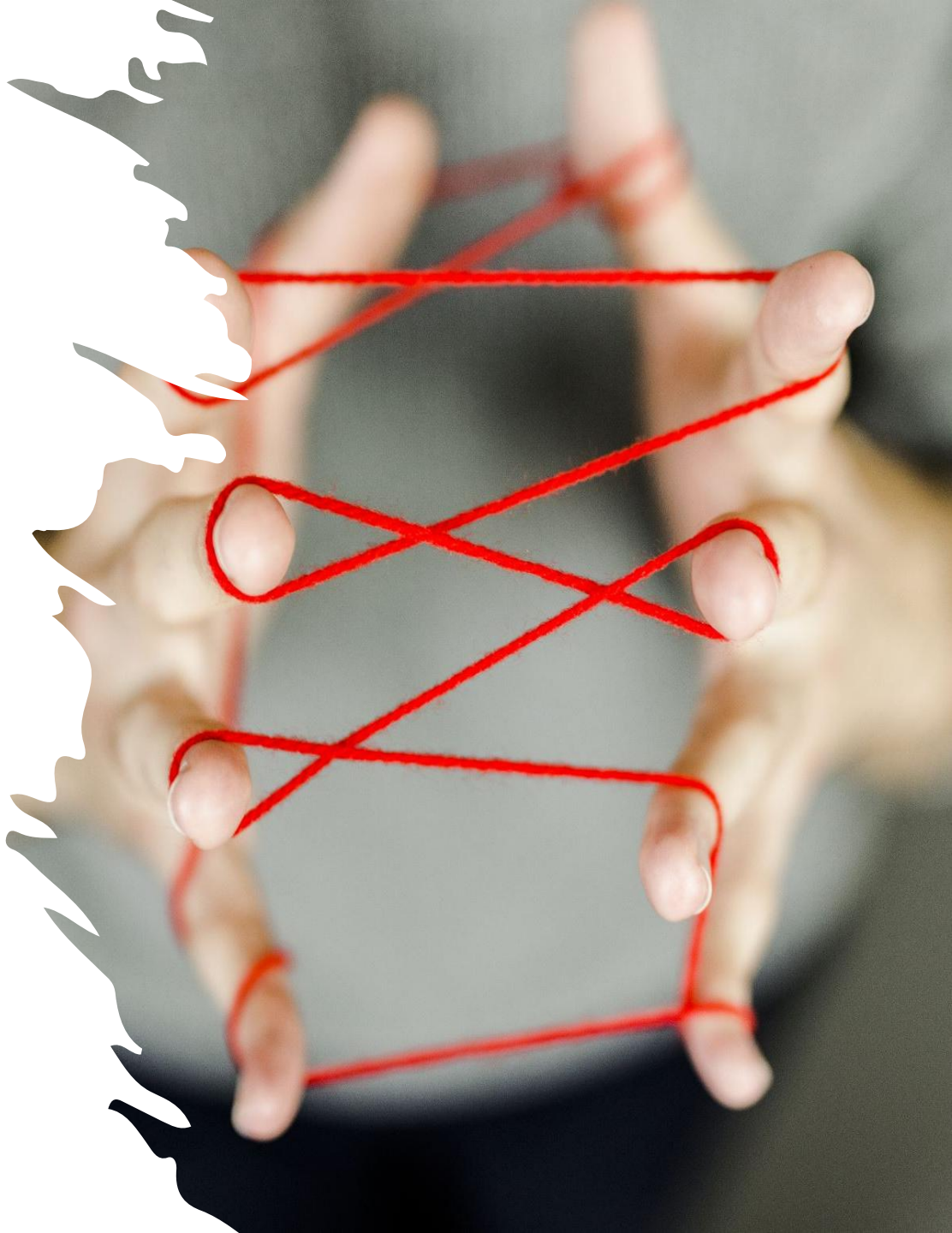
Assoc. Professor of Computational Intelligence

University of Anbar

a.j.aljaaf@uoanbar.edu.iq

Contents

- What is Machine Learning?
- Types of Machine Learning Algorithms.
- Key Differences (Classification vs. Clustering).
- Supervised and Unsupervised Learning.
- Model Training and Testing.
- Fitting a straight Line.
- Prediction Using the Fitted Line.
- Simple Linear Regression.



What is Machine Learning?

Machine learning is a subset of artificial intelligence that focuses on the development of algorithms capable of learning patterns from data and making predictions or decisions without explicit programming. Following are some examples that illustrate its practical applications in various domains.

- One common application of machine learning is **speech recognition**. For instance, when you speak into your mobile phone and mention the name of a contact, some phones can recognize the spoken name and automatically place a call to the intended person. This capability, driven by machine learning, enables the device to process and interpret spoken language effectively.

- Another example is **object recognition**. Suppose you provide an image of an object, such as a car or an airplane, and the system automatically identifies the object. This is achieved through machine learning models trained to recognize patterns in visual data.

What is Machine Learning?

- In **healthcare**, machine learning can be employed to address critical challenges, such as predicting which patients require urgent care. For example, given a limited number of hospital beds, predictive models built using physiological data can identify which patients should be prioritized for emergency care.
- In the **financial sector**, machine learning can help predict customer behaviour. For instance, using customer attributes such as demographics, gender, account balance, and transaction history, a machine learning model can predict the likelihood of a customer defaulting on a loan.

Underlying Framework

At its core, machine learning involves building models that learn from data. These models vary depending on the type of data and the problem being addressed. For example, models can be designed to predict:

- Whether a customer is likely to default.
- The presence of specific objects in an image.
- The content of spoken language in audio recordings.

This process falls within the broader domain of predictive analytics, where machine learning methods are applied to solve a diverse set of problems.

Types of Machine Learning Algorithms

Regression

Classification

Clustering

Regression

Regression refers to predictive modelling tasks where the output is a continuous numerical value. For example, consider a scenario where we aim to predict the final score of a student in a predictive analytics course based on their past academic performance, such as their class 10 marks. Using historical data from previous cohorts, a regression model can be built to predict a student's score, which can range from 0 to 100. Since the target variable is continuous and numeric, this type of problem falls under regression.

Classification

Classification involves predicting discrete categories or classes. A classic example is the spam vs. ham email classification problem, where the goal is to determine whether a particular email is spam or not (ham). In this case, the output is a categorical variable (spam or ham). To build such a model, a large dataset containing emails and their corresponding labels (spam or ham) is required. A machine learning algorithm then learns patterns in the data to classify new, unlabelled emails as either spam or ham.

While regression predicts continuous outcomes, classification predicts categorical ones, and the labels for the target variable are provided during the training process.

Clustering

Clustering is an unsupervised learning approach used to group data into clusters based on similarity, without the need for predefined labels. A well-known example is **market segmentation**, where customer data is analysed to group customers into different segments. For instance, a business might use clustering to identify distinct customer groups and then provide tailored offers, discounts, or product recommendations for each group. Unlike classification, clustering does not start with predefined labels; instead, patterns in the data are discovered during the process.

Another example of clustering can be seen in the domain of news articles. By running a clustering algorithm on a large collection of articles, we can group them into clusters. Upon analysing the clusters, we might find that one group contains articles about current events, another about sports, and so on. The key insight here is that these groupings are discovered rather than predefined.

Key Differences (Classification vs. Clustering)

Although both classification and clustering may appear to deal with categorical outputs, their fundamental difference lies in the presence or absence of labels:

- **Classification:** Labels are predefined. For instance, in the spam vs. ham problem, the dataset includes labelled examples (spam or ham), and the model is trained to classify new data based on these labels.
- **Clustering:** Labels are not predefined. The process begins with raw data, and the clusters are discovered during the analysis. For example, in market segmentation, the clusters represent customer groups that emerge from the data and are later interpreted by analysing their characteristics.

Supervised and Unsupervised Learning

- Having discussed the differences between classification and clustering, it is important to highlight the similarities and differences between regression and classification, as both fall under the category of supervised learning methods. These methods rely on labelled data, where each data point has an associated "tag" or "label" used to train a predictive model. In contrast, clustering is an example of an unsupervised learning method, which works without labels and aims to discover underlying patterns or groupings in the data.

Supervised Learning: Regression and Classification

Regression: In regression, the label is a continuous or numerical value. For example, when predicting a student's score in a Predictive Analytics course, the model learns from historical data that includes the students' previous academic performance (e.g., class 10 marks) and their corresponding course scores. This score, a continuous value between 0 and 100, serves as the "tag" used to train the model.

Classification: In classification, the label is categorical. A classic example is the **spam vs. ham** email classification problem, where the data consists of emails and their corresponding labels indicating whether they are spam or not. The model learns patterns from labelled training data (e.g., spam emails containing words like "lottery" or "drugs") to classify new, unseen emails.

Both regression and classification share the foundational process of supervised learning, where labelled training data is used to build a model, which can later be applied to predict labels for new, unseen test data.

Unsupervised Learning: Clustering

Unlike supervised methods, clustering does not require labelled data. Instead, it groups data points into clusters based on inherent similarities. For example, in market segmentation, clustering algorithms analyse customer data to identify distinct groups, enabling businesses to tailor offers or promotions for each segment. Similarly, clustering can group news articles into categories, such as sports or current events, without predefined labels. The key distinction here is that clusters are discovered from the data, rather than being predefined as in supervised methods.

Model Training (Supervised Learning)

A predictive model is built using a portion of the data called the training dataset, which contains both the input features and the corresponding labels.

- **In regression,** the training dataset might include students' class 10 scores and their Predictive Analytics course scores.
- **In classification,** the training dataset might include 8,000 labelled emails (e.g., spam or ham). The model learns to associate certain patterns, such as the presence of specific keywords, with the labels.

Model Testing (Supervised Learning)

After training, the model is evaluated on a separate test dataset to predict the labels of unseen data points.

- **In regression,** the test dataset might include new students' class 10 scores, and the model is used to predict their course scores.
- **In classification,** the test dataset might include 2,000 new emails, and the model predicts whether each email is spam or ham.

Simple Linear Regression

- In regression analysis, the equation for the regression line is typically written as:

$$Y=\beta_0+\beta_1 X$$

- Here, β_1 is referred to as the **slope parameter**, and β_0 is the **intercept parameter**.

Thank you for listening.

Any questions?